STANDARD OPERATING PROCEDURE (SOP) FOR 3D C-ARM IN SPINAL THEATRES

Post holder responsible for Procedural Document	General Superintendent Radiographer
Author of Standard Operating Procedure	Theatre Specialist Radiographer
Division/ Department responsible for Procedural Document	Specialist Services/ Medical Imaging
Contact details	X3901
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Approving body and date approved	Medical Imaging Specialty Governance Group, 10/10/2023
Review date (and frequency of further reviews)	April 2028 (Every 5 years, 6 months prior to expiry)
Expiry date	October 2028
Date document becomes live	10/10/2023

Please *specify* standard/criterion numbers and tick ✓ other boxes as appropriate

Monitoring Information		**Strategic Directions – Key Milestones**	
Patient Experience	✓	Maintain Operational Service Delivery	✓
Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:			
Note: This document has been assessed for any equality, diversity or human rights implications			

Controlled document
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Full History		Status: Final	
Version	Date	Author (Title not name)	Reason
1.0	10/08/2023	Theatre Specialist Radiographer & General Superintendent Radiographer	New SOP

Associated Trust Policies/ Procedural documents:	*SOP for Performing an X-Ray Examination (RAGN0006), SOP for Theatre and Portable Radiography (RAGN00049)*
Key Words:	*Spinal Surgery, Navigation, 3D imaging, 3D spin, Brain Lab*
In consultation with and date:	
Contact for Review:	Theatre Specialist Radiographer

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KEY POINTS OF THIS PROCEDURAL DOCUMENT

Surgeon/Theatre Co-Ordinator/Surgical Care Practitioner advises X-Ray Superintendent & Theatre Specialist Radiographers of case via email and requests imaging procedure on EPIC

↓

Appropriately trained Radiographer identified on rota in advance of case

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Radiographer attends correct event on CRIS and books into 3D c-arm

↓

Ensure machine is clean and ready for use prior to being taken into theatre environment

↓

Follow departmental ID and pregnancy checking policy

↓

Perform imaging ensuring radiation safety is maintained. X-ray lights, used all staff within theatre during spin must wear 0.35mm lead protection.

↓

Clean machines after use in line with infection control protocols and return to appropriate location

↓

Annotate images and send through to PACS

↓

Post process on CRIS, recording dose (and screening time for theatre)

↓

Check images are correctly delivered to PACS.

1. INTRODUCTION

1.1 This SOP details the correct procedure for performing theatre imaging with the Brain lab navigation system using the 3D C-Arm. This imaging is currently approved for use on adult patients only.

2. PURPOSE

2.1 This document exists to provide guidance to Radiographers and Theatre Staff (including Surgeons, Anaesthetists, Nurses, Surgical Care Practitioners and ODP's).

2.2 This SOP outlines the process for performing the procedures specified and the responsibilities of operators.

3. DEFINITIONS

3.1 Brain Lab – Software system used to provide 3D navigation for surgical accuracy

3.2 EPR – Electronic Patient Record – Currently EPIC

3.3 CRIS – Computer Radiology Information System

3.4 PACs – Picture Archiving System

4. DUTIES AND RESPONSIBILITIES OF STAFF

4.1 Operators are responsible for reading and understanding the SOP and for adhering to the processes for obtaining diagnostic images as outlined in the SOP and also within the SOP for Theatre and Portable Radiography (RAGN00049).

4.2 Operators are responsible for ensuring that correct processes are followed, as outlined in the document, to ensure radiation doses to patients and staff are as low as reasonably achievable.

4.2 General Superintendent Radiographer is responsible for ensuring that all operators are given access to the SOP and a core team of experienced personnel are suitably trained in the use of the machine.

4.3 General Superintendent Radiographer is responsible for ensuring that SOP is shared with all relevant theatre staff via HUB.

4.4 Medical Imaging Engineers and Medical Physics Experts are responsible for ensuring the equipment is maintained and is kept in good working order.

5. STANDARD OPERATING PROCEDURE

5.1 PROCESS FOR ARRANGING IMAGING COVER

5.1.1 Requests for Imaging for 3D navigation cases must be emailed by either the Surgeon, Theatre Co-ordinator or Surgical Care Practitioner directly to the General Superintendent and the Theatre Specialist Radiographers with a minimum of 72 hours (3 working days) notice to enable rostering of suitably trained Radiographers to theatre.

5.1.2 Confirmation that a suitably trained Radiographer is available to cover the navigation list will be emailed back to the Surgeon and the Theatre Co-ordinator.

5.1.3 General Superintendent will roster a Radiographer trained on the navigation system in advance of the list. Wherever possible, 2 suitably trained Radiographers will be made available to cover any 3D navigation lists due to the complexity and safe use of the equipment.

5.2 PROCEDURE FOR IMAGING

5.2.1 Electronic imaging request to be submitted onto EPR in advance of the theatre case with sufficient clinical information to justify the request and radiation dose.

5.2.2 Theatre will be informed of which Radiographer they will have for the case, and what bleep number they are using at the start of the working day. Theatre must bleep the Radiographer before the start of the case. Radiographer will attend imaging request on CRIS before attending theatre.

5.2.3 Radiographer is responsible for ensuring that X-Ray warning lights are operational and sufficient lead protection is available for use.

5.2.4 The surgical ‘Time Out’ can be used to check correct patient identification, operation and side. If the radiographer is not present for surgical ‘Time Out’ then patient identification must be checked with theatre staff and against the consent form.

5.2.5 Pregnancy status (where appropriate) must be checked with the theatre staff / anaesthetist. If a patient has had a pregnancy test, the result should be available on Epic. The Radiographer must then record these details on CRIS.

5.2.6 Upon arrival in theatre Radiographer is to ensure that patient is positioned correctly on theatre table and draped in surgical covers sufficiently to prevent snagging on C-Arm as it completes its rotation.

5.2.7 Radiographer must ensure that they are wearing a lead apron with a minimum lead equivalent of 0.35mm with a corresponding thyroid shield.

5.2.8 Radiographer positions C-Arm with confirmation of AP & Lateral images. When Surgeon is happy with initial position Radiographer conducts a collision check test-spin to ensure that the C-Arm navigates around the patient without any collisions with other equipment, the patient or the theatre table.

5.2.9 When collision check test-spin is confirmed as successful Radiographer must ensure all staff except for the Anaesthetist leave theatre in line with guidance from Radiation

Protection Advisor. The Anaesthetist will then do their final checks to ensure stability of patient condition prior to initiating breath hold for 1 minute, then will leave theatre.

5.2.10 To optimise 3D image quality for any patient over 120kg the increased dose programme must be selected.

5.2.11 Once all non-Radiographic staff have vacated theatre, Radiographer must stand behind mobile lead screen before undertaking 3D imaging.

If Anaesthetist deems patient too unstable to be left they may remain within the theatre only if they are wearing a lead apron with a minimum lead equivalent of 0.35mm with a corresponding thyroid shield and are stood behind the mobile lead screen with the Radiographer.

5.2.12 Upon completion, Radiographer will give visual and audible confirmation of successful imaging and allow Surgeon, Anaesthetist and other theatre staff to return.

5.2.13 Radiographer will return to the Medical Imaging department but will remain available via bleep should the Surgeon require any 2D imaging during the case.

5.3 PROCEDURE POST-OPERATIVELY

5.3.1 Radiographer must send all 2D and 3D images taken during the case as well as the doe report to PACs.

5.3.2 Dose and screening time is to be noted for recording on CRIS.

5.3.3 Examination is to be post-processed on CRIS as per the protocol within the SOP for Performing an X-Ray Examination.

6. ARCHIVING ARRANGEMENTS

6.1 The original of this SOP will remain with the author as listed on the front sheet, (please see page one). This document will be stored electronically on Q-Pulse and HUB. Archived electronic copies will be stored in the obsolete register of Q-Pulse. All paper copies past the review date will be destroyed with only the electronic obsolete copy available on Q-Pulse.

7. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE/ GUIDELINE

7.1 To evidence compliance with this SOP, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Electronic distribution of SOP	Q-Pulse document record	Quarterly distribution acknowledgement reported at the departmental audit

		meeting by the Quality and Accreditation Coordinator
Electronic acknowledgment	Q-Pulse report	Quarterly distribution acknowledgement reported at the departmental audit meeting by the Quality and Accreditation Coordinator
Modality to select at least two SOPs to be audited in each financial year	Audit report	Departmental audit meeting. Non-compliant audits will be noted through the Diagnostics Specialty Governance Group (SGG) meeting via the audit action plan

APPENDIX 1

ISOKERMA MAP FOR ZIEHM VISION 3D SCAN

Measuring results for 3D measurement for FD 12" flat-panel detector

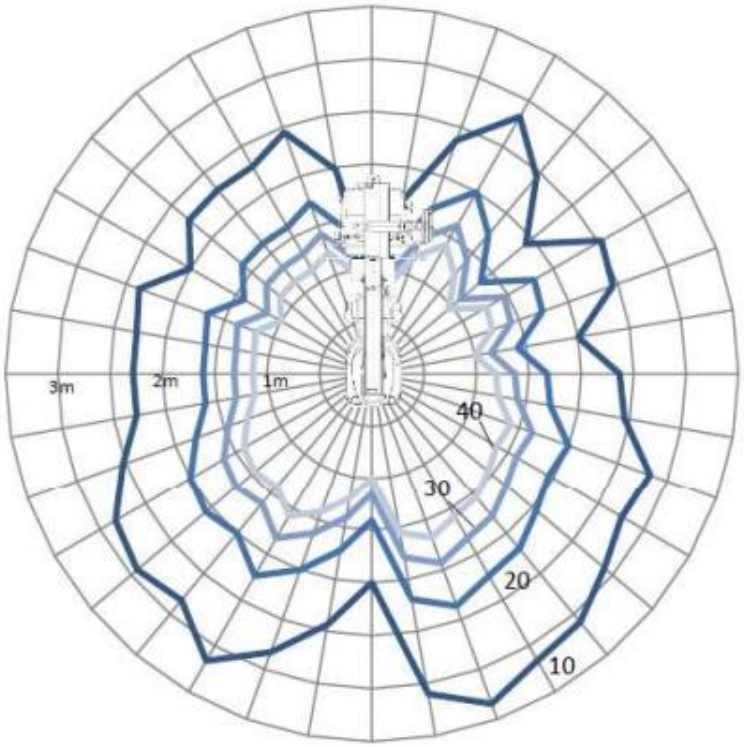

Fig. 61: 3D isokerma maps in µGy for 3D scan, at the height of the isocenter

APPENDIX 2

LIST OF SUITABLY TRAINED RADIOGRAPHERS

Adrian Perry (Theatre Specialist Radiographer)
Natalya Lukyanova (Theatre Specialist Radiographer)
Borja Garcia (Locum Radiographer)
Milly Davies (Radiographer)

APPENDIX 3: COMMUNICATION PLAN

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the guideline/SOP	General Radiographers, Theatre Staff including; Spinal Surgeons, Anaesthetists, Theatre nurses, Surgical Care Practitioners, ODP's
The key changes if a revised document	None
The key objectives	This document exists to provide guidance to Radiographers and Theatre Staff (including Surgeons, Anaesthetists, Nurses, Surgical Care Practitioners and ODP's). This SOP outlines the process for performing the procedures specified and the responsibilities of operators.
How new staff will be made aware of the procedure/guideline and manager action	General Radiographers will have the policy issued for reading during induction to the Trust. To discuss with Theatre as to best process for their staff
Specific Issues to be raised with staff	None
Training available to staff	Specific training and supervision will be provided to a core team of experienced General Radiographers to provide the service using the equipment
Any other requirements	None
Issues following Equality Impact Assessment (if any)	None
Location of hard / electronic copy of the document etc.	SOP to be stored on Q Pulse in Medical Imaging department and on HUB for non-Medical Imaging personnel

APPENDIX 4: EQUALITY IMPACT ASSESSMENT TOOL

Name of document	SOP for 3D C-Arm in Spinal Theatres
Division/Directorate and service area	Specialist Services/ Medical Imaging
Name, job title and contact details of person completing the assessment	Suzanne McLaughlan, General Superintendent/ Lead Reporting Radiographer
Date completed:	08/09/2023

The purpose of this tool is to:

- **identify** the equality issues related to a policy, procedure or strategy
- **summarise the work done** during the development of the document to reduce negative impacts or to maximise benefit
- **highlight unresolved issues** with the policy/procedure/strategy which cannot be removed but which will be monitored, and set out how this will be done.

1. What is the main purpose of this document?

To provide guidance to Radiographers and Theatre Staff (including Surgeons, Anaesthetists, Nurses, Surgical Care Practitioners and ODP's).

To outline the process for performing the procedures specified and the responsibilities of operators.

2. Who does it mainly affect? *(Please insert an "x" as appropriate:)*

Carers ☐ Staff ☒ Patients ☐ Other (please specify)

3. Who might the policy have a 'differential' effect on, considering the "protected characteristics" below? (By *differential* we mean, for example that a policy may have a noticeably more positive or negative impact on a particular group e.g. it may be more beneficial for women than for men)

Please insert an "x" in the appropriate box *(x)*

Protected characteristic	Relevant	Not relevant
Age	☐	☒
Disability	☐	☒
Sex - *including: Transgender, and Pregnancy / Maternity*	☐	☒

Race	☐	☒
Religion / belief	☐	☒
Sexual orientation – *including: Marriage / Civil Partnership*	☐	☒

4. Apart from those with protected characteristics, which other groups in society might this document be particularly relevant to… (e.g. those affected by homelessness, bariatric patients, end of life patients, those with carers etc.)?

None

5. Do you think the document meets our human rights obligations? ☒

Feel free to expand on any human rights considerations in question 6 below.

A quick guide to human rights:
• ***Fairness –*** *how have you made sure it treats everyone justly?* • ***Respect –*** *how have you made sure it respects everyone as a person?* • ***Equality –*** *how does it give everyone an equal chance to get whatever it is offering?* • ***Dignity –*** *have you made sure it treats everyone with dignity?* • ***Autonomy –*** *Does it enable people to make decisions for themselves?*

6. Looking back at questions 3, 4 and 5, can you summarise what has been done during the production of this document and your consultation process to support our equality / human rights / inclusion commitments?

Document written by General Superintendent Radiographer and Theatre Specialist Radiographer in consultation with Radiation Protection Advisor, Radiology Services Manager and the Imaging Quality & Accreditation Facilitator. Consideration given to inclusivity with respects to adhering to IRMER 2018 and the Departmental policy for checking the pregnancy status of individuals having an x-ray procedure.

7. If you have noted any 'missed opportunities', or perhaps noted that there remains some concern about a potentially negative impact please note this below and how this will be monitored/addressed.

"Protected characteristic":	*(e.g age, sex, race, religion)*
Issue:	*The missed opportunity noted*
How is this going to be monitored/ addressed in the future:	
Group that will be responsible for ensuring this carried out:	